

V3 parameter estimation

One model · SPY, AAPL, MSFT, AMZN · four windows · 1.5-year monthly rolling

What this report is

- The monthly rolling parameter paths used in the 1.5-year study, summarized by regime. One table per model.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At each month-end (plus the first trading day of the regime), parameters use only returns and listed quotes with date $\leq$ that update.
- Lookback = 18 months. Rolling = monthly. Same four windows as the decision study.
- Cells are Mean, SD, Min, and Max of those monthly estimates inside the regime.

Tables

- P1 GBM (μ, σ). P2 GARCH ($\mu, \omega, \alpha, \beta, \sigma_0$). P3 Heston ($\mu, \kappa, \theta, \xi, \rho, \nu_0$).
- P4 Merton ($\mu, \sigma, \lambda, \mu_J, \sigma_J, \kappa$). P5 GARCH-Merton. P6 Heston-Merton.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT, AMZN.

Table P1 · Modified GBM · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n=18)																
P(U U)	0.4394	0.0177	0.4176	0.4707	0.4833	0.0208	0.4605	0.5123	0.4593	0.0227	0.4315	0.5027	0.4115	0.0124	0.3866	0.4335
P(D D)	0.4131	0.0148	0.3835	0.4312	0.4394	0.0232	0.3899	0.4601	0.5237	0.0105	0.5077	0.5368	0.4731	0.0172	0.4395	0.4951
P(U D)	0.5869	0.0148	0.5688	0.6165	0.5606	0.0232	0.5399	0.6101	0.4763	0.0105	0.4632	0.4923	0.5269	0.0172	0.5049	0.5605
P(D U)	0.5606	0.0177	0.5293	0.5824	0.5167	0.0208	0.4877	0.5395	0.5407	0.0227	0.4973	0.5685	0.5885	0.0124	0.5665	0.6134
μ_U	0.0127	0.0030	0.0077	0.0158	0.0230	0.0021	0.0190	0.0249	0.0188	0.0037	0.0127	0.0225	0.0282	0.0031	0.0230	0.0315
σ_U	0.0143	0.0039	0.0073	0.0172	0.0193	0.0025	0.0147	0.0213	0.0192	0.0042	0.0118	0.0226	0.0286	0.0010	0.0275	0.0311
μ_D	0.0155	0.0034	0.0095	0.0188	0.0253	0.0027	0.0200	0.0275	0.0182	0.0033	0.0128	0.0215	0.0251	0.0033	0.0191	0.0277
σ_D	0.0156	0.0037	0.0082	0.0183	0.0234	0.0023	0.0182	0.0251	0.0173	0.0035	0.0111	0.0206	0.0231	0.0032	0.0168	0.0256
R2 Normal (n=18)																
P(U U)	0.5706	0.0068	0.5588	0.5841	0.4959	0.0354	0.4514	0.5361	0.5133	0.0111	0.4921	0.5270	0.5041	0.0102	0.4894	0.5256
P(D D)	0.4138	0.0186	0.3854	0.4488	0.4606	0.0173	0.4357	0.4810	0.4482	0.0130	0.4331	0.4724	0.4704	0.0185	0.4423	0.4947
P(U D)	0.5862	0.0186	0.5512	0.6146	0.5394	0.0173	0.5190	0.5643	0.5518	0.0130	0.5276	0.5669	0.5296	0.0185	0.5053	0.5577
P(D U)	0.4294	0.0068	0.4159	0.4412	0.5041	0.0354	0.4639	0.5486	0.4867	0.0111	0.4730	0.5079	0.4959	0.0102	0.4744	0.5106
μ_U	0.0053	1.798e-04	0.0050	0.0055	0.0120	0.0010	0.0106	0.0132	0.0107	2.180e-04	0.0103	0.0110	0.0133	3.760e-04	0.0128	0.0139
σ_U	0.0042	2.958e-04	0.0038	0.0047	0.0106	5.961e-04	0.0099	0.0119	0.0099	2.375e-04	0.0095	0.0102	0.0116	3.218e-04	0.0112	0.0125
μ_D	0.0056	1.571e-04	0.0053	0.0059	0.0114	0.0018	0.0091	0.0136	0.0098	3.190e-04	0.0094	0.0103	0.0127	7.464e-04	0.0111	0.0135
σ_D	0.0051	1.210e-04	0.0049	0.0053	0.0126	0.0022	0.0099	0.0149	0.0116	1.343e-04	0.0114	0.0118	0.0147	0.0020	0.0100	0.0166
R3 Late-cy (n=18)																
P(U U)	0.5616	0.0074	0.5505	0.5724	0.5468	0.0109	0.5251	0.5613	0.5147	0.0103	0.5000	0.5326	0.5949	0.0133	0.5697	0.6149
P(D D)	0.4588	0.0250	0.4018	0.4942	0.4791	0.0106	0.4622	0.4944	0.3564	0.0093	0.3406	0.3719	0.4643	0.0132	0.4391	0.4879
P(U D)	0.5412	0.0250	0.5058	0.5982	0.5209	0.0106	0.5056	0.5378	0.6436	0.0093	0.6281	0.6594	0.5357	0.0132	0.5121	0.5609
P(D U)	0.4384	0.0074	0.4276	0.4495	0.4532	0.0109	0.4387	0.4749	0.4853	0.0103	0.4674	0.5000	0.4051	0.0133	0.3851	0.4303
μ_U	0.0059	6.826e-04	0.0045	0.0066	0.0117	8.063e-04	0.0100	0.0126	0.0108	6.992e-04	0.0092	0.0115	0.0133	9.293e-04	0.0110	0.0140
σ_U	0.0057	6.881e-04	0.0042	0.0062	0.0112	9.124e-04	0.0095	0.0120	0.0107	7.615e-04	0.0090	0.0115	0.0138	0.0012	0.0112	0.0153
μ_D	0.0066	8.330e-04	0.0047	0.0073	0.0122	0.0015	0.0088	0.0135	0.0115	0.0012	0.0087	0.0127	0.0146	0.0017	0.0104	0.0160
σ_D	0.0078	6.019e-04	0.0063	0.0082	0.0130	0.0020	0.0086	0.0147	0.0115	8.387e-04	0.0093	0.0122	0.0153	0.0016	0.0106	0.0165
R4 COVID (n=18)																
P(U U)	0.5619	0.0170	0.5390	0.5920	0.5440	0.0200	0.5197	0.5764	0.5435	0.0157	0.5186	0.5691	0.5445	0.0213	0.5026	0.5738
P(D D)	0.4425	0.0435	0.3703	0.4856	0.4342	0.0323	0.3811	0.4829	0.3817	0.0219	0.3494	0.4156	0.4673	0.0120	0.4494	0.4918
P(U D)	0.5575	0.0435	0.5144	0.6297	0.5658	0.0323	0.5171	0.6189	0.6183	0.0219	0.5844	0.6506	0.5327	0.0120	0.5082	0.5506
P(D U)	0.4381	0.0170	0.4080	0.4610	0.4560	0.0200	0.4236	0.4803	0.4565	0.0157	0.4309	0.4814	0.4555	0.0213	0.4262	0.4974
μ_U	0.0075	0.0015	0.0060	0.0092	0.0140	0.0016	0.0124	0.0164	0.0121	0.0015	0.0105	0.0140	0.0135	7.612e-04	0.0125	0.0150
σ_U	0.0090	0.0035	0.0058	0.0131	0.0137	0.0027	0.0108	0.0169	0.0129	0.0034	0.0097	0.0168	0.0139	8.781e-04	0.0130	0.0157
μ_D	0.0089	0.0023	0.0067	0.0114	0.0144	0.0013	0.0130	0.0167	0.0133	0.0018	0.0113	0.0154	0.0142	8.547e-04	0.0130	0.0157
σ_D	0.0120	0.0049	0.0075	0.0174	0.0172	0.0027	0.0146	0.0206	0.0150	0.0040	0.0110	0.0192	0.0148	8.454e-04	0.0133	0.0163

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling Modified GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are P(U|U), P(D|D), P(U|D), P(D|U), μ_U , σ_U , μ_D , σ_D . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.